

INSTRUCTION LEVEL Parallelisme niveau assembleur

LOAD/STORE Architecture

→ Seuls Load/Store

Accèdent la mémoire

R
I
S
C

 educed
Instruction
et
omporter } $\neq X86$

LOAD source destination
load Mem1, R₂ (Mem1) → R₂

Store

Store R₅, Mem 2. R₅ → (Mem2)

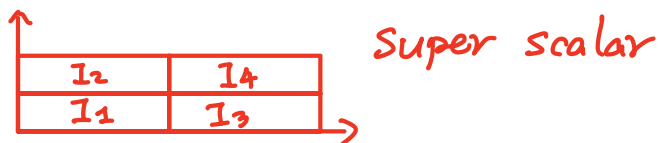
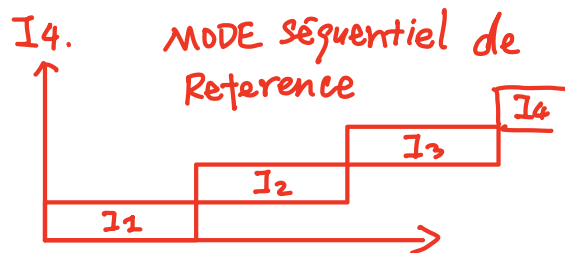
ADD

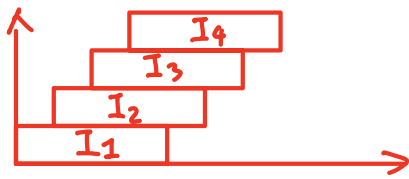
ADD FP1, FP2, FP6 FP1 + FP2 → FP6

Branch

BG2 R₂, (TARGET) IF(R₂) > 0, GO TO TARGET.

BRANCH UNCONDITIONAL
I₁
I₂
I₃
I₄.
CA22
TOTO
TOTO
RETURN





	LOAD	STORE	ADD	BC2
IF 2ms	Charger instruction Calcul	"		"
ID 1ms	Changement registre	"		CR
EX 5	Calcul adresse	"	ADD	Calcul target/eval cond
Mem 3	Accès mémoire	"	X	X
WB 2	Ecrit registre destinat°	X	ecrit reg dest	IF (COND) target → PC

write back

Max : 5 ms.

$A > B$?

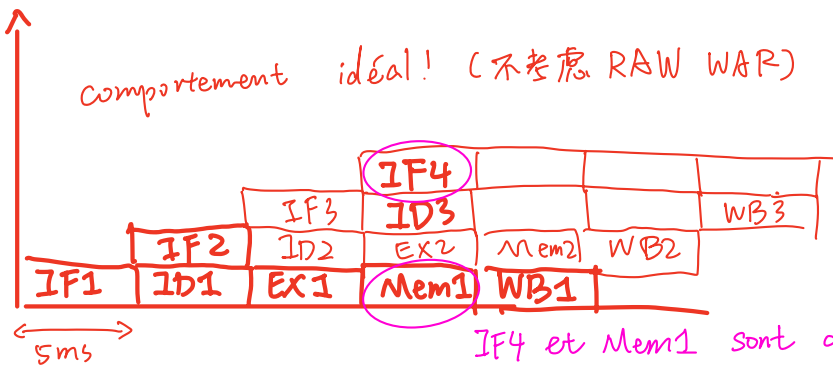
$\Updownarrow$ $B \neq 0$

$A - B$

$A > 0$

	0	1	2	3	4
IF	I1	I2	I3	I4	I5
ID		I1	I2	I3	I4
EX			I1	I2	I3
Mem				I1	I2
WB					I1

← 5ms →



comportement idéal! (ne pas considérer RAW WAR)

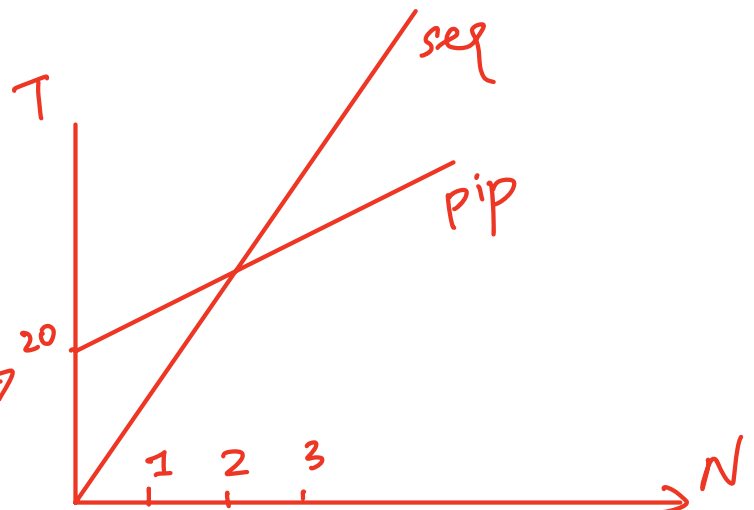
IF4 et Mem1 sont deux accès mémoire

0 1 2 3 4 5 6
5 + 5 + 5 + 5 + 5

	Sequentiel	pipeline
1	13	25
2	26	30
3	39	35

$$T_{seq}(N) = 13 * N \text{ ms}$$

$$T_{pip}(N) = 20 + 5 * N \text{ ms.}$$

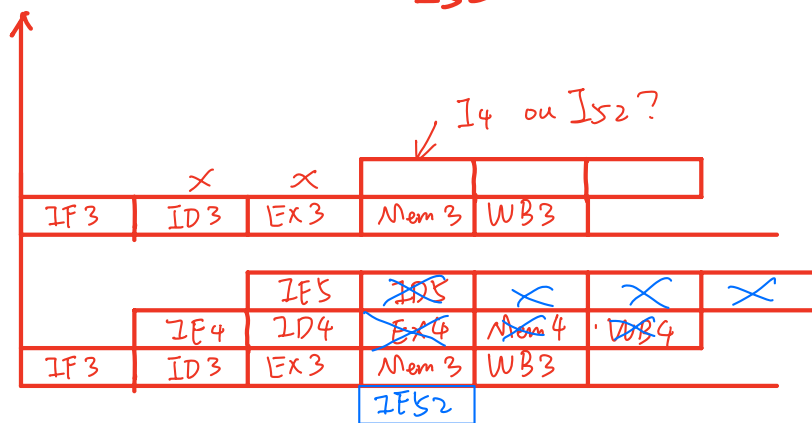


Problème :

→ conflit de ressources (exp. accès de mémoire)

→ BG2: I₃: BG2 I₅₂

I₃ → I₄
→ I₅₂



?? 執行了 IF4, ID4, 但不改變狀態

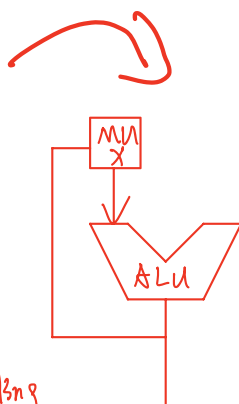
I ₁ \ I ₂	R	W
R	RAR	⚠
W	⚠	⚠

Ressource de stockage partagée entre I₁ et I₂

I₁ ARITHM → REG

		R ₂			
R ₁ + R ₂ → R ₃	IF	ID	EX	Mem	WB
R ₂ * R ₇ → R ₆		IF	ID	EX	Mem

	0	1	2	3	4	5
R ₁ + R ₂ → R ₄	IF	ID	EX	Mem	WB	
R ₄ * R ₅ → R ₆		IF	ID	EX	Mem	



	0	1	2	3	4	5
LOAD R ₄	IF	ID	EX	Mem	WB	
R ₄ * R ₅ → R ₆		IF	ID		EX	Mem

Forwarding

mais perde de cycle

	0	1	2	3	4	5
$R_1 + R_2 \rightarrow R_4$	IF	ID	EX	Mem	WB	
$R_7 * R_4 \rightarrow R_2$		IF	ID	EX	Mem	WB

← R_2

	0	1	2	3	4	5
$R_1 + R_2 \rightarrow R_4$	IF	ID	EX	Mem	WB	
$R_7 * R_8 \rightarrow R_4$		IF	ID	EX	Mem	WB

← R_4

≡
14
≡
12

Contrainte :

→ : 5 PHASE pour tout le monde.

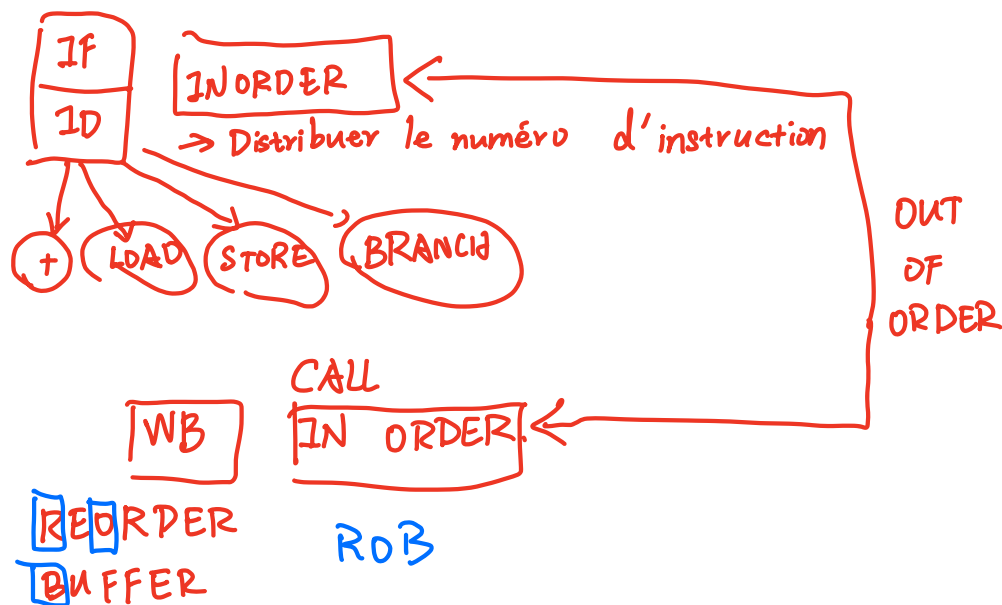
→ 1 Instruction bloquée In order
toutes les suivant bloquées.

→ tout les étape même durée

IF ID EX
mem WB

Amélioration de pipeline. ① forwarding

② ↓



Complex

CISC.: x86

Instruction.

INST complex.

Set

↓ décodeur

Computer

MOD CRISC).

ADD Mem1, R₂, R₃

↓
LOAD Mem1, Tmp → Register.

ADD Tmp, R₂, R₃

RISC : ARM ; Risc V (open source).

DO I = 1, 100,

C[I] = A[I] + B[I]

ENDDO

$A(1:100) \cap C(1:100) = \emptyset$

$B(1:100) \cap C(1,100) = \emptyset$

LOAD @A[I], FP1 A[I] → FP1

INL @A[I]

@A[1] @A[2]

↓
A[1] | A[2] | A[3]

1) code pseudo-code ASM équivalent ?

2) Exécuter en parallèle ?

3) Exécuter pipeline ?

Register renaming

virtuel (16) physical (64)

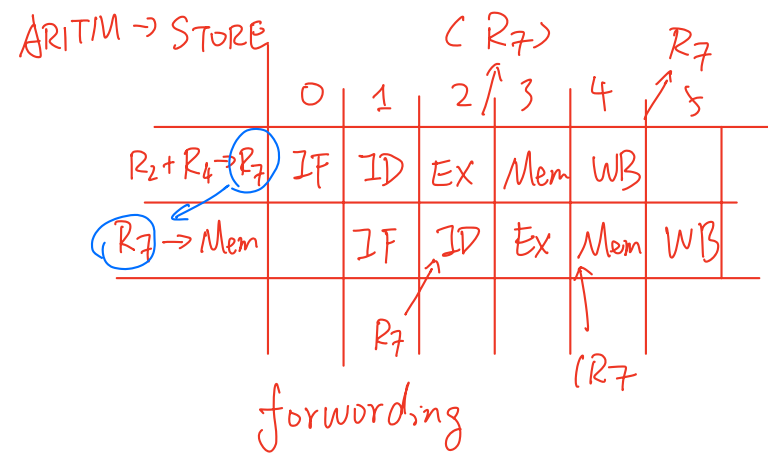
XMM0 → PMM3
XMM1 → PMM4 } IT1

XMM0 → PMM5
XMM1 → PMM8 } IT2

BRANCH : PARI
 shadow register

↗ Taken

↘ not taken



LOAD Mem 1

↓

LOAD Mem 2

ADRESSE LOAD(R₂) 256, R₇

C(REG) OFFSET [(R₂) + 256] → R₇

LOAD Mem 1 R₇

STORE R₈ Mem 2

Mem 1 ? Mem 2

Mem 1 (R₆) 32

Mem 2 (R₈) 12

(R₆) + 32 = (R₈) + 12

Comparaison complete : CHER!

" sur 10 bits Poids faibles (Mem 1)

LOAD	Mem 1, R ₂	IF	ID	EX	Mem	WB
STORE	R ₇ , Mem 1		IF	ID	EX	Mem, WB.

WAP

STORE	R ₇ , Mem 1	IF	ID	EX	Mem	WB
LOAD	Mem 1, R ₈		IF	ID	EX	Mem, WB

MOV R₇, R₈.

Annotations: $(\text{Mem } 1)$ is forwarded from the EX stage of the first instruction to the IF stage of the second instruction. $(\text{Mem } 1)$ is also forwarded from the EX stage of the second instruction to the IF stage of the third instruction.

CLASSIFICATION AP

1960: Flynn

DATA STREAM $\begin{cases} \nearrow \text{SINGLE} \\ \searrow \text{MULTIPLE} \end{cases}$

INSTRUCTION STREAM $\begin{cases} \nearrow \text{SINGLE} \\ \searrow \text{Multiple} \end{cases}$

$\begin{smallmatrix} I \\ D \end{smallmatrix}$	S	M
S	SISD	MISD pas un cas utile
M	<u>SIMD</u>	<u>MIMD</u>

exmple: pipeline

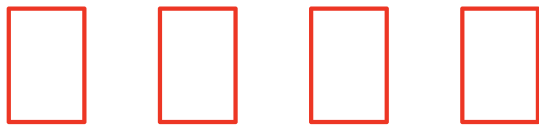
DO I = 1, 100 $\{2, 4\}$ STRIDE.
 $C(I) = A(I) + B(I)$

ENDDO

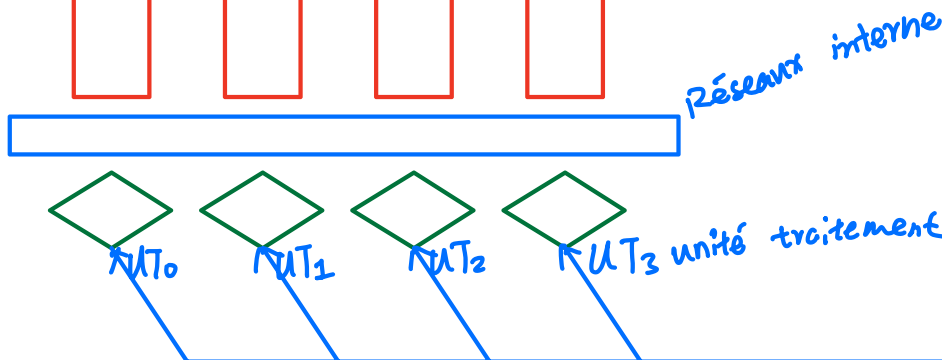
$A(1:100) \cap C(1:100) = \emptyset$

$B(1:100) \cap C(1:200) = \emptyset$

BM_1 BM_2 BM_3 BM_4 bande mémoire.



100 IT $\rightarrow$ 25 tranches de 4 iter
 $\left. \begin{array}{l} 1, 2, 3, 4 \rightarrow TR1 \\ 5, 6, 7, 8 \rightarrow TR2 \end{array} \right\}$



unité de contrôle

LOAD A(), R1

LOAD B(), R2

ADD R1, R2, R3

STORE R3, C()

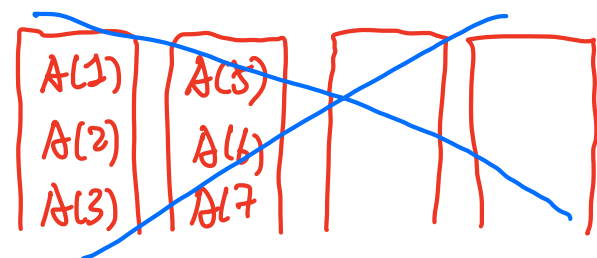
INC @A, @B, @C

UC	UT0	UT1	UT2	UT3
LOAD	A(1) LOCK STEP NODE	A(2) LOCK STEP NODE	A(3) LOCK STEP NODE	A(4)
LOAD	B(1)	B(2)	B(3)	B(4)
ADD				
STORE	C(1)	C(2)	C(3)	C(4)
INCE	+4	+4	+4	+4
Branch	X	X	X	X
LOAD	A(5)	A(6)	A(7)	A(8)
LOAD	B(5)	B(6)	B(7)	B(8)
ADD				
STORE	C(5)	C(6)	C(7)	C(8)

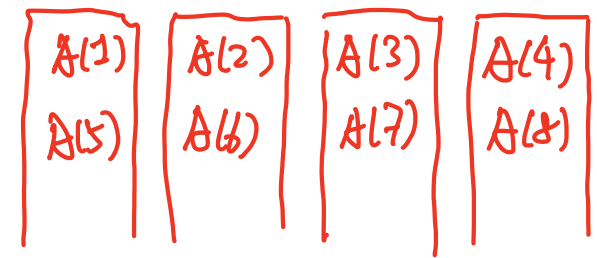
LOCK STEP NODE

TR1

TR2



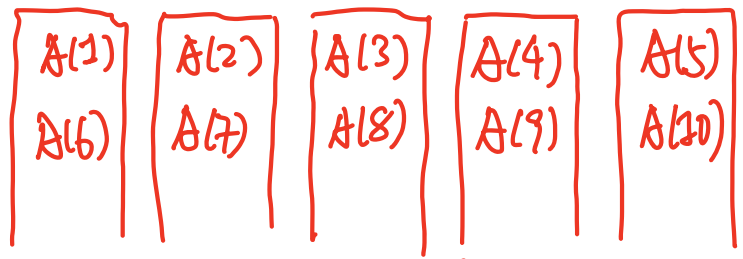
façon parallèle



$A(1) \rightarrow B() \quad K/K \equiv 1-1 \quad [4]$

Cyclique

STRIDE $2 \times S \Rightarrow 4 \text{ mots/cycle} \rightarrow 0.5$
 $2 \mid S \} \Rightarrow 2 \text{ mots/cycle} \rightarrow 0.25$
 $4 \times S \}$
 $4 \mid S \Rightarrow 1 \text{ mots/cycle} \rightarrow 0.25$



$A(1) \rightarrow B() \quad K/K \equiv 1-5 \quad [5]$

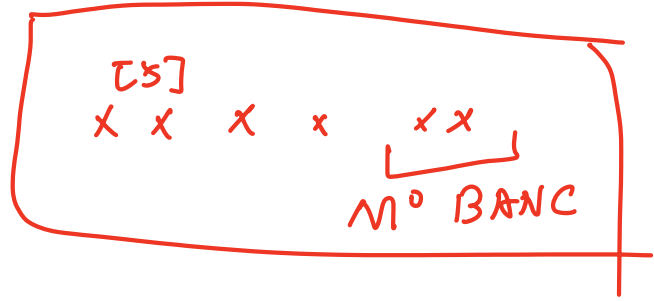
$5 \mid S \Rightarrow 1 \text{ mot/cycle} \quad [0.2 \mid P]$

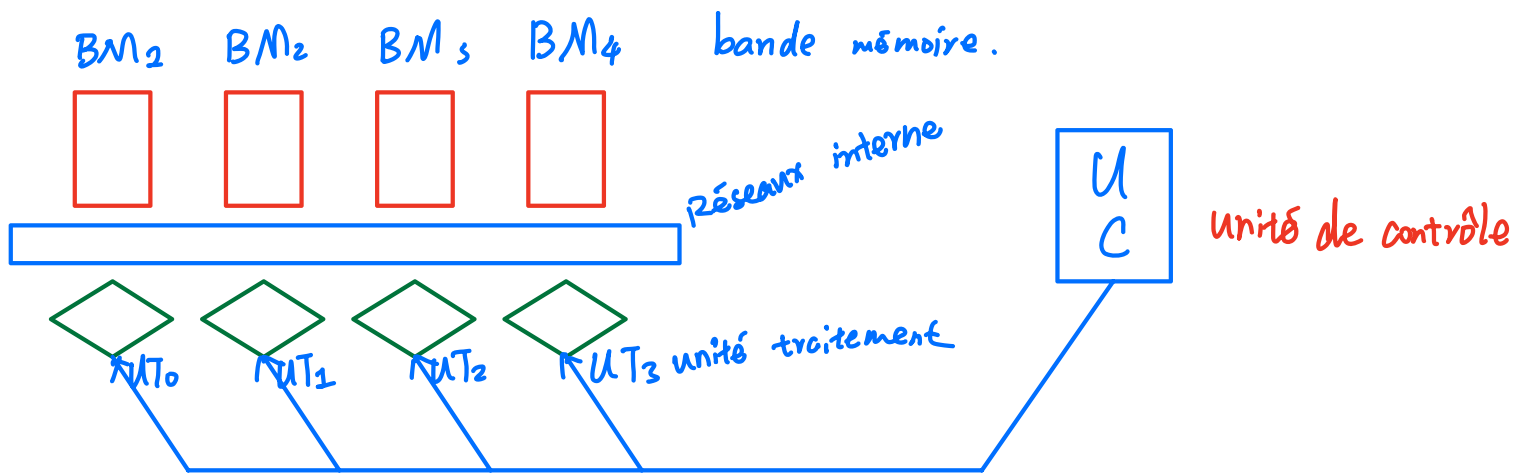
$5 \times S \Rightarrow 5 \text{ mots/cycle} \quad [0.8 \mid P]$

$S=1 \Rightarrow P=0.8$

$S=2 \Rightarrow P=0.1$

$S \neq 1, S \neq 2 \Rightarrow P=0.1$





DO I = 1, 100

C(I) = A(I) + B(I)

ENDDO

$A(I) \rightarrow BM \cdot K /$

$K \equiv I - 1 \quad [4]$

$A(I, J) \quad \left\{ \begin{array}{l} 1 \leq I \leq 4 \\ 1 \leq J \leq 4 \end{array} \right.$

accès par
ligne
D colonne
O diagonale

0	1	2	3
A(1,1)	.. 1,2	1,3	1,4
2,1	2,2	2,3	2,4
3,1	3,2	3,3	3,4
4,1	4,2	4,3	4,4

$|i-j| = n. \quad [4] \quad \downarrow$

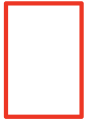
0	1	2	3
A(1,1)	.. 1,2	1,3	1,4
2,4	2,1	2,2	2,3
3,3	3,4	3,1	3,2
4,2	4,3	4,4	4,1

$A(I, J) \rightarrow BM \cdot K /$

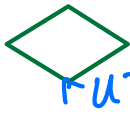
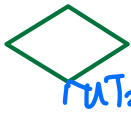
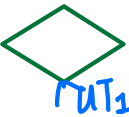
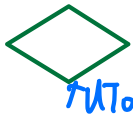
$K \equiv I + J - 2 \quad [4]$

BM₁BM₂BM₃BM₄

bande mémoire.



unité de contrôle

UT₀UT₁UT₂UT₃

unité traitement

IB

IB

IB

IB

DO I = 1, 100

C(I) = A(I) + B(I)

ENDDO

IB = 0

⇒ NO OPERATION

IB = 1

EXEC

STD.

DO I = 1, 100

IF = (A(I) > 0)

THEN T1

ELSE T2

ENDIF

itérat A

(Tranche)

1	>	T ₁
2	>	T ₁
3	>	T ₂
4	<	T ₂
5	<	T ₂
6	<	T ₂
7	>	T ₁
8	>	T ₁

1, 2, 3, 4 → TR 1

5, 6, 7, 8 → TR 2.

UC	UT0	UT1	UT2	UT3
LOAD	A(1)	A(2)	A(3)	A(4)
CMP	>	>	>	<
CMP → IB	IB=1	IB=1	IB=1	IB=0
T_1	T_1	T_1	T_1	X
INV IB	IB=0	IB=0	IB=0	IB=1
T_2	X	X	X	T_2
Set IB	IB=1	IB=1	IB=1	IB=1
LOAD	A(5)	A(6)	A(7)	A(8)
CMP	<	<	>	>
CMP → IB	IB=0	IB=0	IB=1	IB=1
T_1	X	X	T_1	T_2
INV IB	IB=1	IB=1	IB=0	IB=0
T_2	T_2	T_2	X	X

initiation byte.

BGZ R1, (52 12)

IF R1 ≥ 0 (52 12) → PC

ELSE 458 → PC

CMP2 R1 → 0001

compliquer comme.

CMPE.

$R_1, R_2, R_3 \Leftrightarrow R_1 - R_2$.

Greater Equal.

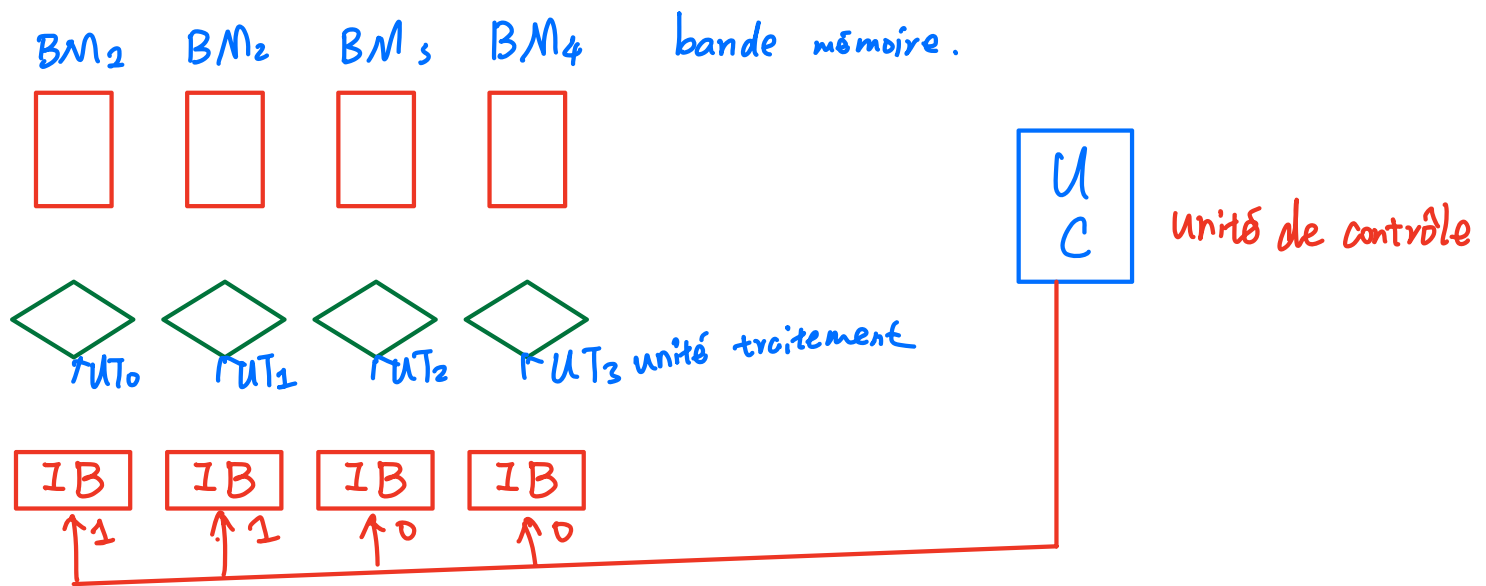
IF $R_1 \geq R_2$ $R_3 = (1, \dots, 1)$

IF $R_1 < R_2$ $R_3 = (0, \dots, 0)$.

>, ≥, <, ≤, =, ≠

=, ≠

Modèle de performance générale



DO $I = 1, N$

$$C(I) = A(I) + B(I) \quad \updownarrow d$$

ENDDU

$T(N)$ division euclidienne

$$N = 4 \times K + R, \quad R \in \{0, 1, 2, 3\}$$

Si $R = 0 \Rightarrow T(N) = K \times d. \Rightarrow$ taux d'utili 25% 50% 75% 100% 625 75 87.5 100 ...

Si $R \neq 0 \Rightarrow T(N) = (K+1) d$

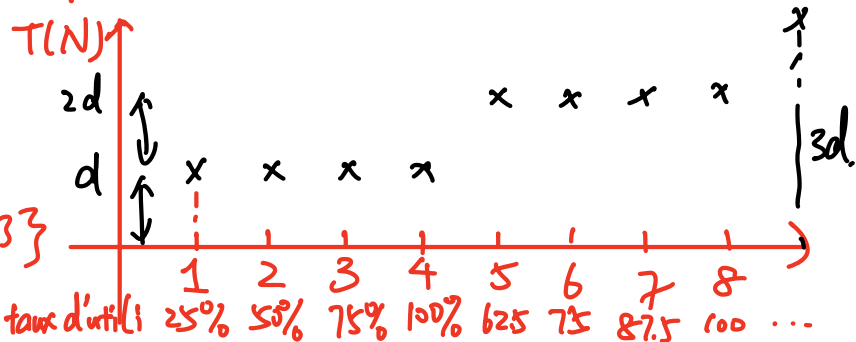
K par rapport à 1

$$K = 1024 \Rightarrow T(4096) = 1024d.$$

$$N = 4096 \Rightarrow 4 \times 1024 = 4096.$$

$$N = 4097 \Rightarrow T(4097) = 1024d + d$$

Temps d'exécution.



$N=1?$
 $N=2$

5 $\rightarrow$ TR complete (1, 2, 3, 4)

TR Incomplete (5)

2048 UT

$$T(N) \quad N \leq 2048 \Rightarrow T(N) = d$$

$$T(2049) = 2d$$

(N Versus nombre d'unité traitement)
N versus # UT.

DO $I = 1, 100$

```

DO I = 1, N
  IF = (A(I) > 0)
    THEN T1 d1
    ELSE T2 d2
  ENDOF
  αN → T1
  (1-α)N → T2

```

α : proba que branchement T soit pris.

Temps total d'exécution .

$$T_1(N) = \alpha \cdot N d_1 + (1-\alpha) N d_2$$

$$= N (\alpha d_1 + (1-\alpha) d_2)$$

SIMD : 4 U.T, 4 IN ⇒ N = 4K.

$$T_4(N) = K (d_1 + d_2)$$

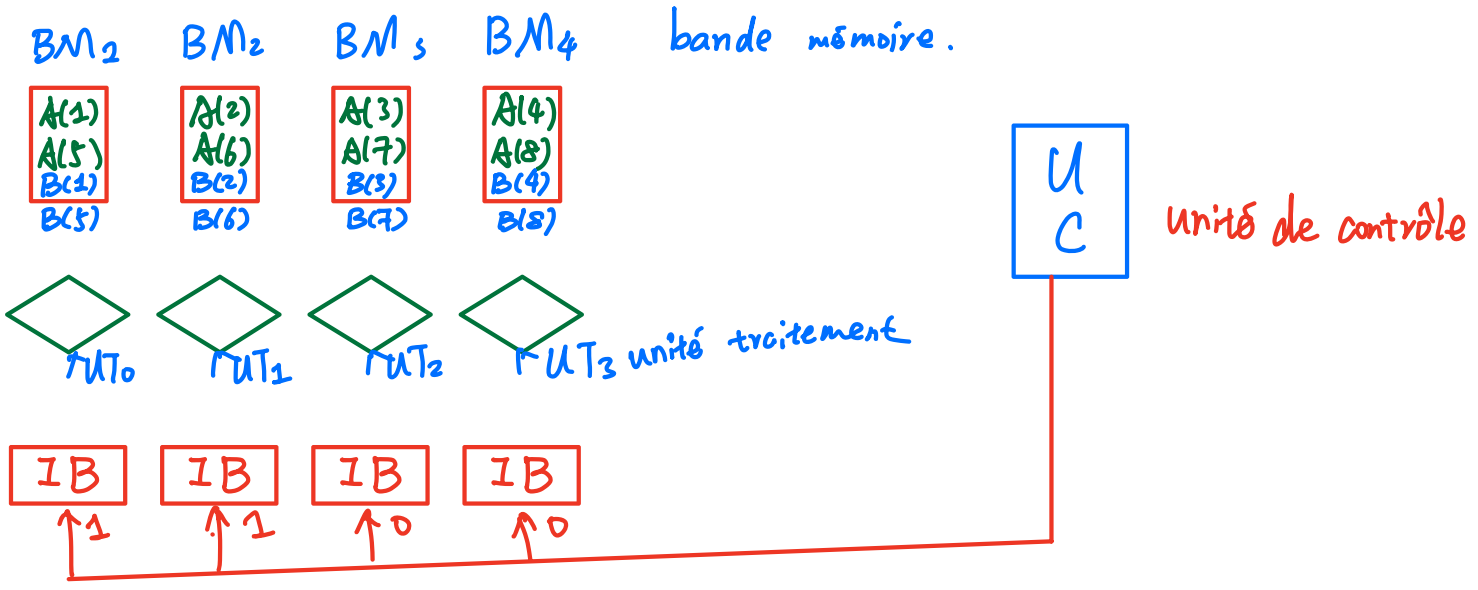
fait ratio :

$$\frac{T_2(N)}{T_4(N)} = \overset{=4}{\left[\frac{N}{K} \right]} \frac{(\alpha d_1 + (1-\alpha) d_2)}{d_1 + d_2}$$

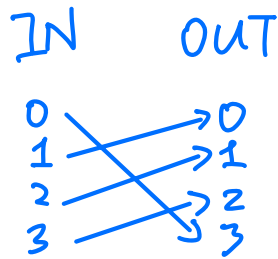
Soit α = 0.5

$\Downarrow$
 0.5

$$\frac{T_1}{T_4} = 2 \quad 2$$



TR1
 $A(1) + B(1)$
 $A(2) + B(2)$
 $A(3) + B(3)$
 $A(4) + B(4)$



DO $I = 1, 100.$ +R
 $C(I) = A(I) + B(I+1)$
 ENDDO
 $A(I) \rightarrow B \wedge K / K \equiv I[4]$

TR1
 $A(1) + B(2)$
 $A(2) + B(3)$
 $A(3) + B(4)$
 $A(4) + B(5)$

IN $I \rightarrow$ OUT $J.$
 $J \equiv I - 1[4]$
 $J \equiv I + P[4]$

- Uniforme.
- parallélisme FP + ACCES
- N nombre itérative.

Versus P
 mbre Ut
 1024 UT
 $N = 1025$
 $N \gg P$

- 1 UC
- Appli // + uniforme
 → graphique
 → IA (matrice multiplication)

CPU : Unité vectorielle.
 8 à 16 UT

AVX 512 512 bits.

8×64 : DP

16×64 : SP

32×64 : FP16

+64b

+32b

+32b

espace logique (adressage)

mémoire par tapée
 (shared memory).

espace physique

réseaux interaction

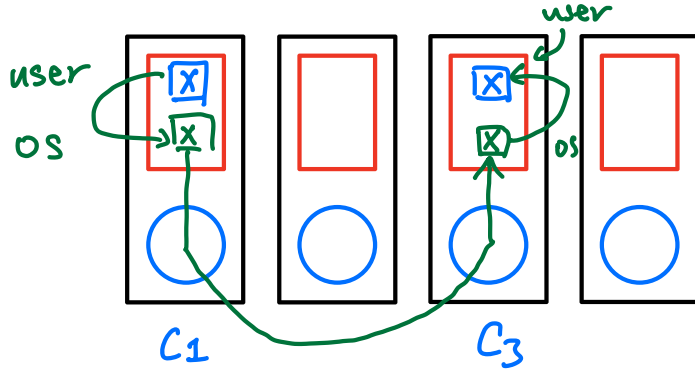
C_1

C_3

wait infini

$\text{Wait}_i(X, C_1)$

$= X$



$X =$
~~SEND~~(X, C₃)
 $X =$
~~RECEIVE~~(X, C₃)
 $X =$
 planter
 Si pas receive.
 mauvaise valeur
 de mise

MESSAGE
PASSING
INTERFACE

DO I = 1, 100

C(I) = A(I) + B(I).

ENDDO

	C ₁	C ₂	C ₃	C ₄
IT	1→25	26→50	51→75	76→100
A	"	"	"	"
B	"	"	"	"
IT	1→10	11→20	21→30	31→40

Machine à mémoire distribuée.

	C ₁	C ₂	C ₃	C ₄
IT	1→25	26→50	51→75	76→100
C	"	"	"	"

IT 1-10 11→20 21→30

send
 (C(11→20), C₂)
 send
 (C(21→25), C₃)
 Recive
 (C(11→20), C₂)
 send
 (C(26→30), C₃)
 Recive
 (C(21→25), C₁)
 send
 (C(31→40), C₄)
 recive
 (C(26→30), C₂)

IT 1→25 26→40.
 send
 C(1→25), C₂
 C₃
 C₄
 recive
 (C(31→40), C₄)

mémoire partagée
 + simple à programmer
 - cout (Réseau)
 - Synchronisat°
 - performance

mémoire distribuée.
 + cout construire
 - programmation. data.
 + perf local
 - perf non-local.

